

UNIT 1 QUESTIONS

Short Answer Questions:

1. Define ADT.
2. Define data structure.
3. What do you mean by non-linear data structure? Give an example.
4. What is a linear data structure? Give an example.
5. List the various operations that can be performed on a data structure.
6. List out the areas in which data structures are applied extensively.
7. How are data structures classified?
8. Differentiate linear and non-linear data structures.
9. What is a linked list?
10. Define a doubly linked list.
11. Define Radix sort and give an example.
12. Difference between arrays and lists.
13. State the advantages of circular lists over a doubly linked list.
14. What is the advantage of a linked list over arrays?
15. What is the circular linked list?
16. Write the common errors on the linked list with examples.

Long Answer Questions:

1. Explain the insertion operation in a linked list. How nodes are inserted after a specified node.
2. Write an algorithm to insert a node at the beginning of the list with an example.
3. What are the applications of linked lists in dynamic storage management?
4. What are the benefits and limitations of a linked list?
5. Define the deletion operation from a linked list.
6. What are the different types of data structures?
7. Explain the operation of traversing a linked list. Write the algorithm and give an example.
8. Explain the operations of singly linked lists with an example.
9. Explain the operations of doubly linked lists with an example.
10. Explain the operations of circularly linked lists with an example.
11. Explain the polynomial ADT with an example of the subtraction operation.

UNIT 2 QUESTIONS

Short Answer Questions:

17. Define Stack.
18. What are the various Operations performed on the Stack?
19. Write the postfix form for the expression $A+B-C+D$?
20. What are the postfix and prefix forms of the expression? $A+B*(C-D)/(P-R)$
21. Explain the usage of a stack in the recursive algorithm implementation?
22. Define Queue.
23. What are the various operations performed on the Queue?
24. How do you test for an empty Queue?
25. Define Circular Queue.
26. List any four applications of a stack.
27. What are the types of queues?
28. What are the applications of a queue?
29. Distinguish between a stack and a queue.

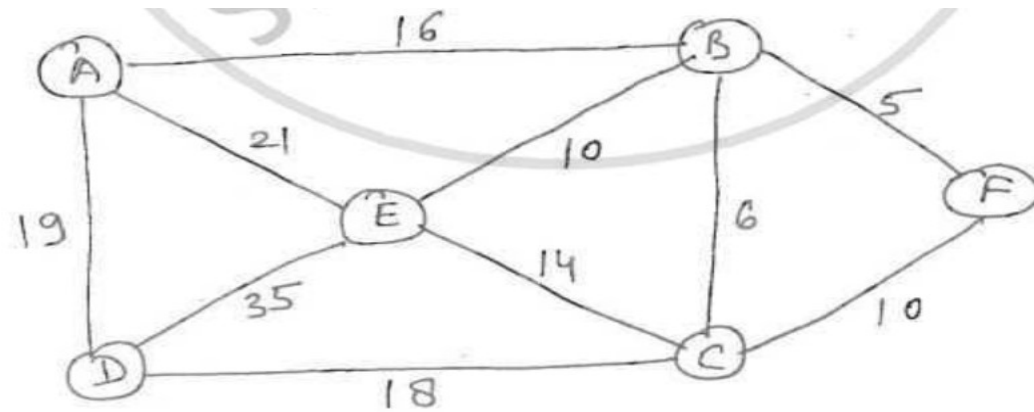
Long Answer Questions:

12. Explain the array implementation of the queue ADT in detail?
13. Explain the addition and deletion operations performed on a circular queue with necessary algorithms.
14. Explain in detail the priority queue ADT?
15. Write the algorithm for converting an infix expression to a postfix (reverse polish) expression and explain with an example?
16. Write the algorithm for converting an infix expression to a prefix (polish) expression and explain with an example?
17. Explain the linked list implementation of the stack ADT in detail?
18. Explain the array-based implementation of stacks.
19. Explain how to evaluate arithmetic expressions using stacks.
20. Explain a circular queue and its implementation
21. Explain a double-ended queue and its operations

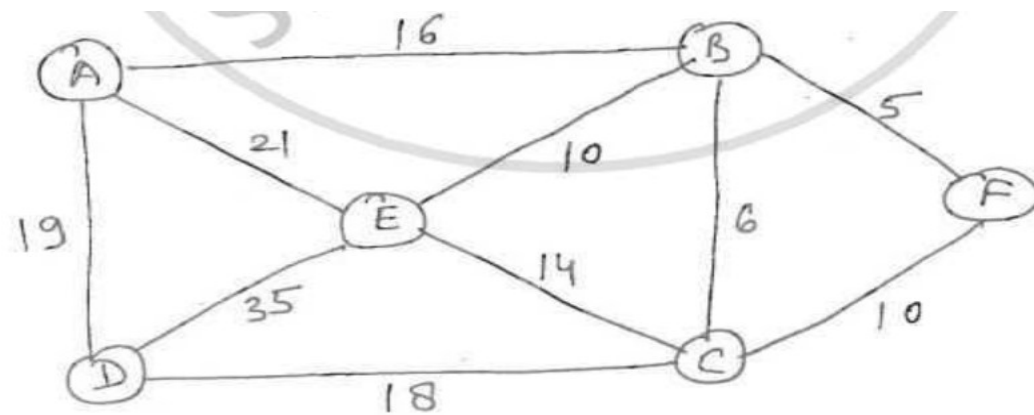
QUESTION BANK

MODULE - 5: Graphs

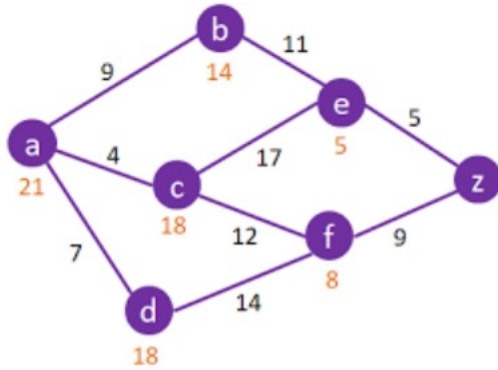
1. Define Graph?
2. What is a directed and undirected graph?
3. Define in degree and out degree of a graph?
4. What is an acyclic graph?
5. Explain the different ways of representing a graph with an example?
6. What is a minimum spanning tree?
7. Define minimum cost spanning tree?
8. Explain the different operation of graph with a function.
9. Explain Dijkstra's algorithm with an example?
10. Explain the Prim's algorithm to find minimal spanning tree for a graph with an example.
11. Explain the Kruskal's algorithm to find minimal spanning tree for a graph.
- 12.
13. Write and explain the BFS algorithm and DFS algorithm with an example.
14. What is Spanning Trees? Explain Spanning Tree in detail with example.
15. Differentiate between depth first search and breadth first search.
16. For the graph given below, find the minimum spanning graph using Prims algorithm



17. For the graph given below, find the minimum spanning graph using Kruskal's algorithm



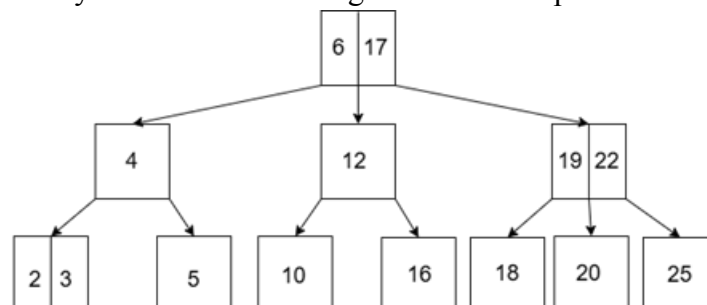
18. For the graph given below, Explain step by step BFS and DFS Graph traversal methods



19. Formulate an algorithm to find the shortest path using Dijkstra's algorithm and explain with example.
20. Explain the minimum spanning tree algorithms with an example.

BST

1. What is the use of Make_null() in BST operations?
2. Differentiate between Find_min() and Find_max() functions.
3. Demonstrate node insertion in a BST containing {25, 15, 35, 10, 20, 30, 40}.
4. Analyze the steps involved in deleting a node with two children from a BST.
5. Develop a complete algorithm to perform all basic BST operations
6. Define an AVL Tree. Why is it called a self-balancing binary search tree?
7. Explain single rotation in an AVL tree.
8. Identify situations where a double rotation is required in AVL trees.
9. Illustrate the need for balancing in AVL trees with an example where a single rotation is required.
10. Compare left rotation, right rotation, left-right rotation, and right-left rotation in AVL trees.
11. Demonstrate the steps involved in performing a Right-Left (RL) double rotation with an example.
12. Construct an AVL tree for the insertion sequence: 10, 20, 30, 40, 50, 25. Show intermediate steps and indicate where single and double rotations occur.
13. Examine all four types of rotations (LL, RR, LR, RL) used in balancing an AVL tree and evaluate their impact on the tree's height.
14. Design examples to show how AVL rotations maintain the balance factor property after insertion and deletion and justify why each rotation is necessary.
15. Define the terms order and height in the context of a B-Tree.
16. What happens when a node in a B-Tree becomes overfull during insertion? Briefly explain the process used to handle it.
17. What is a B-Tree? Mention one key property that differentiates it from a Binary Search Tree.
18. Construct a B-Tree of order 3 for the following keys:
10, 20, 5, 6, 12, 30, 7, 17.
Show the tree after each insertion.
19. Delete the key 4 from the following B-Tree and explain the resultant tree with diagram.



20. Differentiate B-tree with Binary search Tree. How searching can be executed in both trees? Explain with example.
21. Construct a B-Tree of order 3 for the sequence of keys:

12, 4, 5, 25, 1, 32, 16, 8, 15, 20

Show the step-by-step insertion process using diagrams.

Then, delete keys 20, and 5, explaining and draw how the tree changes after each deletion.

22. Explain all the structural properties of B-Trees in detail:

- a. Order and degree
- b. Height and balance
- c. Node capacity (min and max keys)
- d. Leaf node characteristics
- e. Splitting and Merging

23. Construct a B-Tree of order 3 and BST for the sequence of keys:

24, 18, 6, 2, 20, 14, 7, 17

Search for the key 20 in both the trees. How the searching operation differs between a Binary Search Tree (BST) and a B-Tree.

24. Construct an Expression Tree for the expression: $(A + B) * (C - D)$

25. What traversal of an expression tree gives the infix expression?

26. Write the prefix and postfix expressions for the infix expression: $(A + B) * (C - D)$

27. What is the advantage of using an Expression Tree?

28. How are operands and operators represented in an expression tree?

29. Discuss how inorder, preorder, and postorder traversals are used in Expression Trees with suitable examples.

30. Write an algorithm to construct an Expression Tree from a prefix expression.

31. Evaluate an Expression Tree for the postfix expression $A B + C D - *$ if $A=2$, $B=3$, $C=5$, $D=4$.

32. Draw an Expression Tree for the infix expression $(A + B) * (C - D)$ and write its prefix and postfix forms

33. Given the Expression Tree below, write its infix, prefix, and postfix expressions:



34. Construct and evaluate an Expression Tree for the infix expression

$(A + B) * (C - D)/E$. Write its corresponding prefix, postfix, and infix expressions and explain each traversal clearly.

35. Explain how Expression Trees are used in compilers for expression evaluation and code generation.

36. Given the postfix expression:

$A B + C D - * E /$

(a) Construct the corresponding Expression Tree.

(b) Write its equivalent infix and prefix expressions.

(c) Explain how the Expression Tree helps evaluate the result when $A=2$, $B=3$, $C=5$, $D=4$, $E=2$.

37. What is preorder traversal used for?

38. If preorder is A, B, D, E, C, F and inorder is D, B, E, A, F, C — what is the root node?

39. In a binary tree, which traversal gives leaf nodes first?

40. Explain inorder, preorder, and postorder traversals with examples.

41. Discuss the use of traversal in expression tree evaluation.

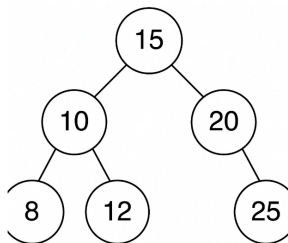
42. Given inorder: D, B, E, A, F, C and preorder: A, B, D, E, C, F — construct the binary tree.

43. Given traversals (Inorder: D, B, E, A, F, C; Postorder: D, E, B, F, C, A), reconstruct the tree and show all traversals.

44. In a binary expression tree, internal nodes are operators and leaves are operands. Given: $A + (B * C) - D$, draw the tree and show all traversals.

45. Write a algorithm to implement inorder, preorder, and postorder traversals of a binary tree.

46. Create a binary tree using arrays and another using linked representation. Compare both by printing their index–node mapping (L2)



47. In a hospital system, patients are categorized by priority for treatment using a Binary Search Tree (BST).

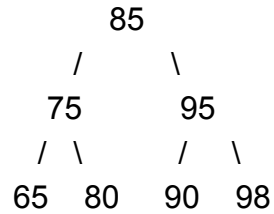
Higher urgency → left side; Lower urgency → right side. (L4)

Build the BST for patient priority levels {50, 30, 70, 20, 40, 60, 80}.

Each node represents a patient (circle with ID/priority).

Operations:

- a. Insert new patient priority.
- b. Perform inorder traversal .
- c. Find patient with highest urgency
- d. Find patient with least urgency



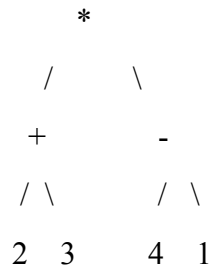
Expected Output:

Inorder: 65 75 80 85 90 95 98

Topper: 98

Lowest: 65

48. A compiler must evaluate the expression $((2 + 3) * (4 - 1))$. (L6)



1. Build the expression tree.
2. Implement postorder traversal to evaluate it.
3. Print each node visited and the result.

49.

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51. .